CBS COPENHAGEN BUSINESS SCHOOL
HANDELSHØJSKOLEN

LOGISTIC REGRESSION

BONUS MATERIAL

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Topics

- Derivation of Cross-entropy loss
- Stochastic gradient descent
- Generative vs. Discriminative Classifier
- Regularization
- Multinomial Logistic Regression

Deriving cross-entropy loss for a single observation x

How far is the classifier output $\hat{y}$ from the true output y?

$J(\hat{y},y)$

- How much $\hat{y} = \sigma(w \cdot x+b)$ differs from true y = 0 or 1

Goal

- Maximize probability of the correct label $p(y|x)$

For the binary outcomes, probability from model: $p(y|x)$

$$p(y|x) = \hat{y}^{y} . (1-\hat{y})^{(1-y)}$$

if y=1 -> $p(y|x) = \hat{y}$

if y=0 -> $p(y|x) = 1-\hat{y}$

Deriving cross-entropy loss for a single observation x

Whatever values maximize log p(y|x) will also maximize p(y|x)

- Take the log of both sides

Maximize
$$\begin{aligned}\log p(y|x) &= \log\left[\hat{y}^{y}\,(1-\hat{y})^{1-y}\right] \\ &= y\log\hat{y} + (1-y)\log(1-\hat{y})\end{aligned}$$

- Now flip sign to turn this into a loss: Minimize Cross-entropy loss

Minimize $J(\hat{y}, y) =$
$$-\log p(y|x) = -\left[y\log\hat{y} + (1-y)\log(1-\hat{y})\right]$$

$$-\left[y\log\sigma(w\cdot x+b) + (1-y)\log\left(1-\sigma(w\cdot x+b)\right)\right]$$

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Gradient Descent: Example

Heart disease prediction based on age, blood pressure and cholesterol level

P(Y=1|x) : Estimated probability of the patient having heart disease	
P(Y=1|x) > 0.5 Patient has heart disease	P(Y=1|x) $\leq$ 0.5 Patient has no heart disease

$b = -1.5$

$w_1 = 0.04$,
Age = 50 years

$w_2 = 0.03$,
bp = 130 mm Hg

$w_3 = 0.02$,
chol = 240 mg/dL

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Gradient Descent: Example

- Predicted output: $\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$

  Where $z = b + age \times w_1 + bp \times w_2 + chol \times w_3$

- Cross Entropy Loss:

$$J(w) = -\left[y \log \sigma(z) + (1-y) \log\left(1 - \sigma(z)\right)\right]$$

- Gradient of Cross Entropy Loss:

$$\frac{\partial J}{\partial b} = \sigma(z)\text{-}y, \quad \frac{\partial J}{\partial w_1} = (\sigma(z)\text{-}y) \times age, \quad \frac{\partial J}{\partial w_2} = (\sigma(z)\text{-}y) \times bp, \quad \frac{\partial J}{\partial w_3} = (\sigma(z)\text{-}y) \times chol$$

- Update Parameter values as:

$$b := b - \alpha \frac{\partial J}{\partial b}, w_j := w_j - \alpha \frac{\partial J}{\partial w_j}$$

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Gradient Descent: Example

Parameter	Initial Value	Gradient $\frac{\partial J}{\partial w_j}$	Updated Value $w_j - \alpha \frac{\partial J}{\partial w_j}$	New Parameter (After 1 Iteration)
b	-1.5	0.5	$-1.5 - 0.01 \times 0.5$	-1.505
w_1	0.04	-0.1	$0.04 - 0.01 \times (-0.1)$	0.041
w_2	0.03	0.2	$0.03 - 0.01 \times 0.2$	0.028
w_3	0.02	-0.05	$0.02 - 0.01 \times (-0.05)$	0.021

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Generative vs. Discriminative Classifier

Generative Classifiers	Discriminative Classifiers
Tries to model class - what are the features of the class.	Learn what features are most useful to distinguish between the various classes.
Learns the joint probability distribution p(x,y) of label x and y. It predicts the conditional probability and then picking a most likely label.	Learns the conditional probability distribution p(y|x).
Example: Naive Bayes	Example: Logistic Regression
$\hat{c} = \underset{c \in C}{\operatorname{argmax}} \overbrace{P(d\|c)}^{\text{likelihood}} \overbrace{P(c)}^{\text{prior}}$	$\hat{c} = \underset{c \in C}{\operatorname{argmax}} \overbrace{P(c\|d)}^{\text{posterior}}$

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Multinomial Logistic Regression

- If output class >2, then multinomial logistic (Softmax) regression
  - Ex: Pos / Neg / Neu
  - P(Positive|doc) + P(Negative|doc) + P(Neutral|doc) = 1
- Need a generalization of the sigmoid called the softmax
  - For each feature we'll need separate weight vectors for each of the K classes
- It takes a vector z = [z_1, z_2, ... , z_k] of k arbitrary class
  - Outputs a probability distribution in range (0,1) and sum of all values equal to 1

$$\text{softmax}(z) = \left[\frac{\exp(z_1)}{\sum_{i=1}^{k} \exp(z_i)}, \frac{\exp(z_2)}{\sum_{i=1}^{k} \exp(z_i)}, \ldots, \frac{\exp(z_k)}{\sum_{i=1}^{k} \exp(z_i)} \right]$$

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Example - Multinomial Logistic Regression

- Raw scores (logits): z=[z_1, z_2, z_3]=[2.0, 1.0, 0.5]
- Exponential of each score: e^z =[e^2, e^1, $e^{0.5}$]

$$\text{Softmax}(z) = \frac{e^{z_1}}{e^{z_1}+e^{z_2}+e^{z_3}}, \frac{e^{z_2}}{e^{z_1}+e^{z_2}+e^{z_3}}, \frac{e^{z_3}}{e^{z_1}+e^{z_2}+e^{z_3}}$$

$$\text{Probability(pos)} = \frac{e^2}{e^2+e^1+e^{0.5}} = 0.629$$

$$\text{softmax}(z) = \left[\frac{\exp(z_1)}{\sum_{i=1}^{k}\exp(z_i)}, \frac{\exp(z_2)}{\sum_{i=1}^{k}\exp(z_i)}, \ldots, \frac{\exp(z_k)}{\sum_{i=1}^{k}\exp(z_i)}\right]$$

$$\text{Probability(neg)} = \frac{e^1}{e^2+e^1+e^{0.5}} = 0.231$$

$$\text{Probability(neu)} = \frac{e^{0.5}}{e^2+e^1+e^{0.5}} = 0.140$$

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